Acyclovir kinetics after intravenous infusion

The disposition and safety of the antiviral drug acyclovir were studied in 14 subjects with advanced malignancies. Acyclovir was administered by a 1-hr intravenous infusion at doses of 0.5, 1.0, 2.5, and 5.0 mg/kg. At the end of infusion, mean peak plasma levels (±SEM), determined by radioimmunoassay, were 6.4 ± 0.7, 12.1 ± 2.3, 14.9 ± 2.7, and 33.7 ± 7.1 μM. The plasma concentration-time profiles could be described by a biexponential equation. The half-life of acyclovir in the slow disposition phase ranged from 2.2 to 5 hr and the drug was detected in the plasma for at least 18 hr after infusion. The total body clearance ranged from 117 to 396 ml/min/1.73 m^2. A proportionality between area under the curve and dose suggests that acyclovir exhibits dose-independent kinetics in the dose range studied. There was wide variation in cumulative urinary excretion of unchanged drug, ranging from 30 to 69% of the dose. From renal clearances of acyclovir, which were higher than creatinine clearances, it appears that both glomerular filtration and tubular secretion contribute to its renal excretion. Analysis of the urine by reverse-phase high-performance liquid chromatography revealed the presence of the metabolite 9-carboxymethoxymethylguanine. There was no indication of toxicity either clinically or from laboratory findings in any of the study subjects. This study demonstrates that in addition to selectivity and low toxicity, the kinetic profile and metabolic disposition of acyclovir make it an attractive candidate for therapy in a variety of herpes infections.

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The acyclic nucleoside of guanine [acyclovir, 9-(2-hydroxyethoxymethyl)guanine]* has been found to be a potent and selective inhibitor of viruses of the herpes group, particularly herpes simplex.[4, 24] Selective inhibition of herpes simplex virus results from anabolism to its monophosphate derivative by a virus-coded

*This compound has previously been called acycloguanosine and BW248U; the proprietary name is Zovirax.

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Table I. *Patient demographic data*

Subj. No.	*Sex*	*Age (yr)*	*Wt (kg)*	*Height (cm)*	*BSA* (m^2)*	*Type of cancer*	*Relevant surgical history*	*Concurrent medication*
1	M	67	61.0	168	1.71	Bladder	Cystectomy and prostatectomy	Cimetidine, oxycodone
2	M	68	81.6	185	2.07	Prostate	—	Oxycodone
3	F	49	41.0	160	1.37	Breast	Mastectomy	Oxycodone, trimethobenzamide
4	M	51	47.0	170	1.49	Bladder	Cystectomy	Acetaminophen, prochlorperazine
5	M	59	75.0	193	2.00	Hodgkins	—	Empirin, digoxin, isosorbide dinitrate
6	F	53	117.0	178	2.43	Breast	Mastectomy	Diazepam, dihydromorphinone
7	M	68	70.0	180	1.88	Hypernephroma	Nephrectomy	Busulfan
8	M	68	74.0	178	1.92	Bladder	—	Oxycodone, diazepam
9	M	62	84.0	185	2.09	Hypernephroma	Nephrectomy	Oxycodone
10	F	49	44.0	160	1.40	Breast	Mastectomy	Acetaminophen, trimethobenzamide
11	M	56	80.0	193	2.07	Hypernephroma	Nephrectomy	Codeine, diazepam, oxycodone, hydroxyzine
12	M	72	69.0	175	1.84	Prostate	Prostatectomy	Diethylstilbestrol
13	F	82	57.0	164	1.62	Bladder	—	Prednisone, flurazepam
14	M	56	67.5	173	1.82	Lung	—	Oxycodone, flurazepam

*BSA = $0.0235\ H^{\cdot 4226}\ W^{\cdot 51456}$.

thymidine kinase and subsequent conversion to the di- and triphosphates, presumably by cellular kinases.[8, 10] The triphosphate of acyclovir has been shown to inhibit viral DNA synthesis.[8] In uninfected cells there is essentially no phosphorylation of the drug and consequently little interference with cellular DNA synthesis. The efficacy of acyclovir has been demonstrated in several types of herpes infections in laboratory animals; herpes keratitis in rabbits,[1, 15, 22, 24] mouse herpes encephalitis,[24] cutaneous herpes infections in guinea pigs,[24] some latent herpes simplex infections in mice,[9] and genital herpes infections in mice and guinea pigs.[16] Its effectiveness in man has been indicated in herpes simplex corneal ulcers.[14]

Studies of acyclovir disposition in mice, rats, dogs, and rhesus monkeys have shown considerable interspecies differences in absorption and metabolism.[5] Kinetic studies in dogs have shown that the plasma half-life ($t_{1/2}$) of acyclovir ranged from 2.2 to 4.1 hr.[17] The beneficial effect of acyclovir on survival and natural progression of disease in animal studies, its lack of toxicity at effective doses,[27] and its limited catabolism have warranted its development in phase I clinical trials in man. Our study was designed to evaluate single-dose kinetics and safety of acyclovir after administration by constant-rate intravenous infusion to subjects with advanced malignancies.

Materials and methods

Patient population. The initial dose-tolerance, safety studies of acyclovir in man were limited to patients with advanced malignancies, refractory to available chemotherapy, and at increased risk of severe herpes virus infection. Ten men and 4 women, ranging in age from 49 to 82 yr and in weight from 41 to 117 kg, were selected as subjects. All patients received concurrent medications during the study period (Table I). None of our patients received either cancer or antiviral chemotherapy within 1 wk and none of the women had child-bearing potential. Complete physical and laboratory evaluations were performed (hematologic, renal, and hepatic evaluation). None of our subjects had severe hepatic or renal dysfunction.

Toxicity monitoring. Serial toxicology studies were performed at weekly intervals and included evaluations of bone marrow, renal, and

liver function, complete blood counts, differential white counts, reticulocyte count, bilirubin, serum glutamic oxaloacetic transaminase, creatinine, blood urea nitrogen, alkaline phosphatase, and total protein. Adverse clinical experiences were sought daily.

Drug administration and sample collections. The lyophilized sodium salt of acyclovir (prepared at the Wellcome Development Laboratories) was reconstituted as a 2.0% solution with sterile water to a final concentration of 20 mg/ml. The calculated dose was added to sterile lactated Ringer's solution sufficient to bring the total volume to 60 ml (pH = 9.1). This 60-ml solution was infused intravenously at a constant rate over a 1 hr period (1 ml/min) into a peripheral vein. Each patient received a single dose of 0.5, 1.0, 2.5, or 5.0 mg/kg (Table III). Venous blood samples (5 ml) were drawn into heparinized tubes from the contralateral arm prior to infusion, and at the following times after beginning the infusion: 0.25, 0.50, 0.75, 1.00 (end of infusion), 1.17, 1.33, 1.67, 2, 3, 5, 7, 11, 19, and 37 hr. The 0.25- and 0.75-hr samples during the infusion were not obtained for patients Nos. 1 to 7. Each blood sample was centrifuged immediately and the separated plasma was frozen for subsequent analysis.

Total urine output was collected for 72 hr in timed fractions. The first sampling was immediately before infusion and then, in the time intervals of 0 to 1 hr (during infusion) 1 to 5 hr, 5 to 9 hr, 9 to 13 hr, 13 to 25 hr, 25 to 48 hr, and 48 to 72 hr.

Radioimmunoassay for acyclovir. Plasma and urine concentrations of acyclovir were determined by a sensitive radioimmunoassay highly specific for acyclovir.[23] The assay is performed in a single tube for each sample by adding to 100 μl plasma or urine, 200 μl phosphate-buffered saline, 100 μl diluted antiserum (obtained from rabbits immunized with a succinyl acyclovir rabbit serum albumin conjugate), and 100 μl succinyl-[^{3}H]acyclovir (3.28 Ci/mmole). Overnight incubation of the assay tubes is followed by the addition of 50 μl bovine gamma globulin (10 mg/ml) and 550 μl cold saturated ammonium sulfate. After centrifugation (2,230 × g) and washing of the pellet with half saturated ammonium sulfate, the final precipitate of the antigen-antibody complex is dissolved by addition of 100 μl deionized water followed by 100 μl 4N HCl and 2.0 ml Aquasol-2 (New England Nuclear) scintillator. A standard curve is constructed with each run of the assay with predose plasma or urine samples, at the same dilution as the unknowns, to which known amounts of acyclovir are added. All samples are run in triplicate. From the scintillation counting data the standard curves are obtained by plotting concentrations vs percent binding (B/Bo) on log-logit paper. Routinely, the standard curve data are subjected to an iterative weighted least-squares regression analysis and the unknowns are determined from the regression equation with a computer program. The lower limit of detection of the assay is 50 pmole/ml (11.3 ng/ml).

High-performance liquid chromatography. Selected diluted urine samples (100 μl), clarified by centrifugation, were injected onto a short precolumn in series with a Waters μBondapak C_{18} column and eluted isocratically at a flow rate of 24 ml/hr with a 25 mM potassium phosphate buffer, pH 3.3, containing 3% ethanol.

Pharmacokinetic analysis. Initial graphical inspection of the postinfusion plasma concentration vs. time data of each patient clearly revealed biexponential decay. Individual patient data were, therefore, analyzed through the use of the nonlinear least-squares regression program NONLIN[20] in conjunction with a specific subroutine for the 2-compartment open kinetic model with zero-order input as expressed by equation 1.

$$C_1 = \frac{ko\,(k_{21} - \alpha)\,(1 - e^{\alpha\tau})\,e^{-\alpha t}}{V_1\alpha\,(\alpha - \beta)} + \frac{ko\,(k_{21} - \beta)\,(1 - e^{\beta\tau})\,e^{-\beta t}}{V_1\beta\,(\beta - \alpha)} \quad (1)$$

where

- ko = rate of infusion
- α = rapid disposition rate constant
- β = slow disposition rate constant
- k_{21} = distribution rate constant out of peripheral compartment
- V_1 = apparent volume of distribution of the central compartment
- τ = duration of infusion at time t
- t = time from beginning of infusion
- C_1 = plasma acyclovir concentration at time t

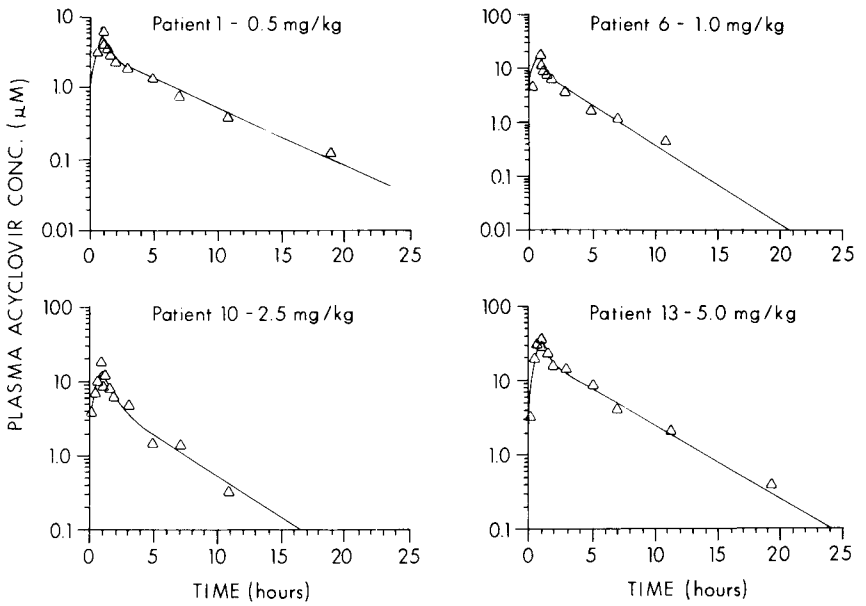


Fig. 1. Plasma concentration time profiles of acyclovir during and following a 1-hr intravenous infusion at various doses. The solid line is the curve calculated from NONLIN and the triangles represent the experimental data points.

Table II. *Average plasma acyclovir levels (mean ± SEM, μM) after various doses*

	Dose			
Time (hr)	*0.5 mg/kg (n = 3)*	*1.0 mg/kg (n = 4)*	*2.5 mg/kg (n = 2)*	*5.0 mg/kg (n = 5)*
0.25	—*	—*	5.63 ± 1.77	3.09 ± 0.86
0.50	2.61 ± 0.42	4.44 ± 1.19	11.3 ± 4.33	14.5 ± 2.33
0.75	—*	—*	14.8 ± 4.10	24.1 ± 3.87
1.00	6.44 ± 0.66	12.1 ± 2.33	14.9 ± 2.67	33.7 ± 7.08
1.17	4.06 ± 0.18	8.60 ± 1.53	11.7 ± 0.24	29.9 ± 4.52
1.33	3.44 ± 0.08	6.75 ± 0.67	9.45 ± 0.69	27.5 ± 2.6
1.67	2.53 ± 0.17	5.36 ± 0.65	8.31 ± 0.32	21.3 ± 1.71
2.00	2.12 ± 0.17	4.58 ± 0.57	6.66 ± 0.35	15.7 ± 1.06
3.00	1.63 ± 0.14	3.16 ± 0.25	4.22 ± 0.51	10.4 ± 2.60
5.00	0.99 ± 0.20	1.91 ± 0.09	1.68 ± 0.15	7.04 ± 1.18
7.00	0.60 ± 0.13	1.18 ± 0.08	1.54 ± 0.19	4.11 ± 0.75
11.00	0.36 ± 0.06	0.41 ± 0.03	0.65 ± 0.28	2.00 ± 0.39
19.00	0.07 ± 0.03	0.09 ± 0.02	0.09 ± 0.03	0.51 ± 0.16
37.00	<0.05	<0.05	<0.05	<0.05

*Specimens not obtained.

This equation describes the plasma concentration of drug both during and after infusion. Parameter estimates were obtained for α, β, k_{21}, and V_1. A weighting function of $1/C_1$ was used in the analysis. The initial estimates for input into NONLIN were obtained from the exponential stripping program CSTRIP,[25] which was modified to calculate estimates of k_{21} and V_1. The rate constant for distribution into the peripheral compartment, k_{12}, the central compartment elimination rate constant, k_{10}, and the apparent volume of distribution at steady state, V_{dss}, were calculated according to the following equations.

$$k_{10} = \alpha\beta/k_{21} \quad (2)$$

$$k_{12} = \alpha + \beta - k_{21} - k_{10} \quad (3)$$

$$V_{dss} = \frac{(k_{12} + k_{21})}{k_{21}} V_1 \quad (4)$$

Table III. *Kinetic parameters*

Subject No.	*Dose (mg/kg)*	*NONLIN derived parameters*							
		α (hr^{-1})	β (hr^{-1})	$t½\ \beta$ (hr)	k_{12} (hr^{-1})	k_{21} (hr^{-1})	k_{10} (hr^{-1})	V_1 (1/1.73 m^2)	Vdss (1/1.73 m^2)
1	0.5	2.50	0.181	3.83	1.31	0.831	0.545	13.3	34.2
2		2.01	0.182	3.81	0.976	0.540	0.680	14.4	40.4
3		4.03	0.322	2.15	2.07	1.13	1.15	8.03	22.7
Mean		2.85	0.228	3.26	1.45	0.833	0.792	11.9	32.4
±SEM		0.61	0.047	0.56	0.32	0.170	0.183	2.0	5.2
4	1.0	0.676	0.186	3.73	0.142	0.423	0.297	32.4	43.2
5		3.40	0.239	2.90	1.79	1.14	0.713	13.2	33.9
6		5.10	0.342	2.03	2.78	1.17	1.49	6.66	22.5
7		3.22	0.242	2.86	1.69	0.822	0.946	11.5	35.2
Mean		3.10	0.252	2.88	1.60	0.889	.862	15.9	33.7
±SEM		0.91	0.033	0.35	0.54	0.174	.249	5.65	4.4
9	2.5	3.48	0.259	2.68	1.83	0.872	1.04	18.0	55.8
10		1.09	0.250	2.77	0.296	0.491	0.557	31.3	50.2
Mean		2.29	0.255	2.73	1.06	0.682	.798	24.7	53.0
±SEM		1.20	0.005	.05	.77	.191	.242	6.7	2.8
11	5.0	—	0.233	2.98	—	—	—	—	—
12		0.806	0.139	4.99	0.275	0.342	0.327	55.7	100.5
13		2.05	0.233	2.98	0.826	0.960	0.496	24.3	45.0
14		4.66	0.206	3.36	2.90	0.971	0.999	10.3	40.9
8		0.94	0.215	3.22	0.163	0.708	0.286	49.3	60.6
Mean		2.11	0.205	3.51	1.04	0.745	0.527	34.9	61.8
±SEM		0.89	0.017	0.38	0.64	0.148	0.164	10.7	13.6
Total mean		2.61	0.231	3.16	1.31	0.800	0.733	22.8	45.0
±SEM		0.42	0.014	0.20	0.27	0.778	0.102	24.4	5.6

Table IV. *Renal clearance*

Subject No.	*Dose (mg/kg)*	*Serum* creatinine (mg/100 ml)*	*Estimated creatinine clearance (ml/min/1.73 m^2)*	*Measured creatinine clearance (ml/min/1.73 m^2)*
1	0.5	1.2	49.6	45.1
2	0.5	1.0	64.7	64.5
3	0.5	0.7	87.9	—
4	1.0	1.1	65.2	—
5	1.0	1.4	55.7	56.0
6	1.0	1.0	55.4§	40.5
7	1.0	1.6	43.4	63.0
9	2.5	1.3	62.2	38.5
10	2.5	1.1	57.2	56.9
11	5.0	1.5	55.5	42.9
12	5.0	1.3	50.8	53.4
13	5.0	1.4	33.1	26.8
14	5.0	1.6	50.0	41.0
8	5.0	1.1	65.2	48.0

*Predose.

†Numbers in parentheses indicate correlation coefficients.

‡Incomplete collection.

§Based on LBW.

Model independent parameters		
Corr	Cl_{TOT} (ml/min/1.73 m²)	AUC_0^∞ (nmole · hr)/ml
0.990	117	19.4
0.955	157	16.1
0.996	161	11.9
	145	15.8
	14	2.2
0.995	150	27.0
0.967	152	31.5
0.999	174	35.4
0.984	177	27.0
	163	30.2
	7	2.0
0.934	311	46.0
0.970	291	37.0
	301	41.5
	10	4.5
—	396	70.6
0.945	304	85.5
0.995	200	115
0.977	160	149
0.929	235	124
	259	109
	42	14
	213	
	21	

Excreted unchanged in 72 hr (%)	Renal clearance (ml/min/1.73 m²)	Cl_R/Cl_{TOT}
69	96.5 (0.994)†	0.823
63	131.1 (0.997)	0.836
68‡	138.2 (0.995)	0.858
−‡	—	—
51	78.6 (0.916)	0.515
66	160.0 (0.986)	0.919
53	115.4 (0.997)	0.652
46	171.7 (0.973)	0.552
64	158.2 (0.955)	0.544
65	559.1 (0.933)	1.413
60	245.4 (0.932)	0.808
39	77.0 (0.890)	0.386
30	48.5 (0.779)	0.304
48	83.1 (0.892)	0.354

Total body clearance (Cl_{TOT}) was calculated by the following model-independent method:

$$Cl_{TOT} = \frac{\text{Dose}}{\int_0^\infty C_t dt} = \frac{\text{Dose}}{AUC_0^\infty} \quad (5)$$

where AUC_0^∞ is the area under the plasma concentration-time curve from time 0 to infinity determined according to the trapezoidal rule. The terminal plasma t½ of acyclovir was calculated with the equation $t½\beta = \ln 2/\beta$.

From the postinfusion data, renal clearance (Cl_R) was determined by linear least-squares regression analysis (no intercept) of the rate of urinary excretion of unchanged drug during a collection interval vs the plasma concentration at the midpoint of this interval. The slope of this regression line is the Cl_R.

Body surface area (BSA) was estimated for each subject from the relationship $BSA = 0.0235H \cdot^{422} \times W \cdot^{515}$ where H is height in centimeters and W is weight in kilograms.[11]

Creatinine clearance (Cl_{cr}) was determined directly from the first 25 hr of fractional urine collections, starting at the time of infusion. To verify the completeness of collection and to provide values for those patients whose urine collection was known to be incomplete, estimates of creatinine clearance were computed based on patient's age, sex, weight (BW), and serum creatinine (S_{cr}) according to the method of Mawer et al.,[19] where for males

$$Cl_{cr} = \frac{BW\ (29.3\text{-}0.203\ age)}{14.4\ S_{cr}}$$

and for females

$$Cl_{cr} = \frac{BW\ (25.3 - 0.175\ age)}{14.4\ S_{cr}}$$

Age is expressed in years, BW in kilograms, and S_{cr} in milligrams per 100 millimeters. When body weight exceeded lean body weight estimates[6] by 10% the latter was used in the calculation of Cl_{cr}. Creatinine clearance estimates were normalized to 1.73 m² BSA. The kinetic parameter values at 0.5, 1.0, and 5.0 mg/kg dose levels were analyzed statistically with Student's t test, and $p < 0.05$ was considered statistically significant. The 2.5 mg/kg dose level was excluded from statistical treatment as there were only 2 subjects in this group.

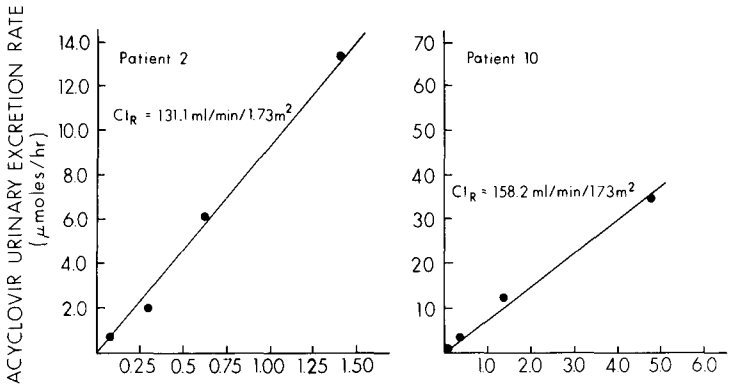


Fig. 2. Urinary excretion rate plotted against the midinterval plasma concentration for 2 patients. The solid line was obtained by regression analysis and the closed circles are data points. The slope of the regression line is the renal clearance.

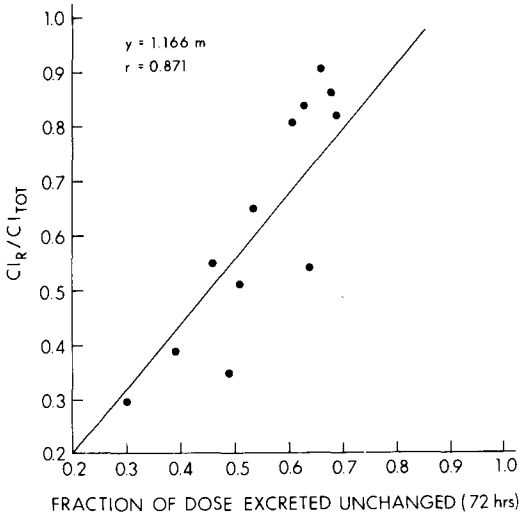


Fig. 3. Plot of Cl_R/Cl_{TOT} against fraction of dose excreted unchanged. The solid line was obtained from the regression analysis and the closed circles correspond to the data points.

Results

Pharmacokinetic analysis. The postinfusion plasma concentrations of acyclovir declined biphasically with the exception of Patient No. 11, the plasma concentration-time data of each patient were well fit by a 2-compartment open model with zero-order input (equation 1). Typical "best fit" curves superimposed over experimental data points are shown in Fig. 1. The arithmetic mean plasma acyclovir concentrations at each dose level during and after infusion are shown in Table II. Mean peak concentration (± SEM) at the end of infusion was 6.4 ± 0.7, 12.1 ± 2.3, 14.9 ± 2.7 and 33.7 ± 7.1 μM, for the 0.5, 1.0, 2.5 and 5.0 mg/kg dose, respectively. Six hr after the end of the infusion mean concentrations were 0.6 ± 0.1, 1.2 ± 0.1, 1.5 ± 0.2 and 4.1 ± 0.8 μM for the same doses.

The resultant kinetic parameter values for each patient are presented and summarized according to dose level in Table III. For Patient 11, model independent parameters were calculated along with the plasma t½ determined by log-linear least-squares regression analysis of the postdistribution plasma concentration-time data and included in Table III. The volume and clearance parameters have been normalized to 1.73 m^2 body surface area. Also presented in this table is the NONLIN reported correlation coefficient (corr) relating the calculated and observed plasma concentrations. The mean t½βs for the 0.5, 1.0, 2.5, and 5.0 mg/kg infusions were 3.26, 2.88, 2.73, and 3.51 hr. The Cl_{TOT} means for these doses were 145, 163, 301, and 259 ml/min/1.73 m^2; for the V_{dss}, means were 32.4, 33.7, 53.0, and 61.8 *l*/1.73 m^2 at the 0.5, 1.0, 2.5, and 5.0 mg/kg doses, and for V_1 (reported in Table III) were approximately 50% of V_{dss}'s. There was no statistically significant difference in any of the kinetic parameters at the dose levels analyzed. There was a proportionality between AUC_0^∞ and dose (r = 0.908).

The renal excretion parameters are summarized in Table IV. The 72-hr recovery of unchanged drug in the urine ranged from 30% to

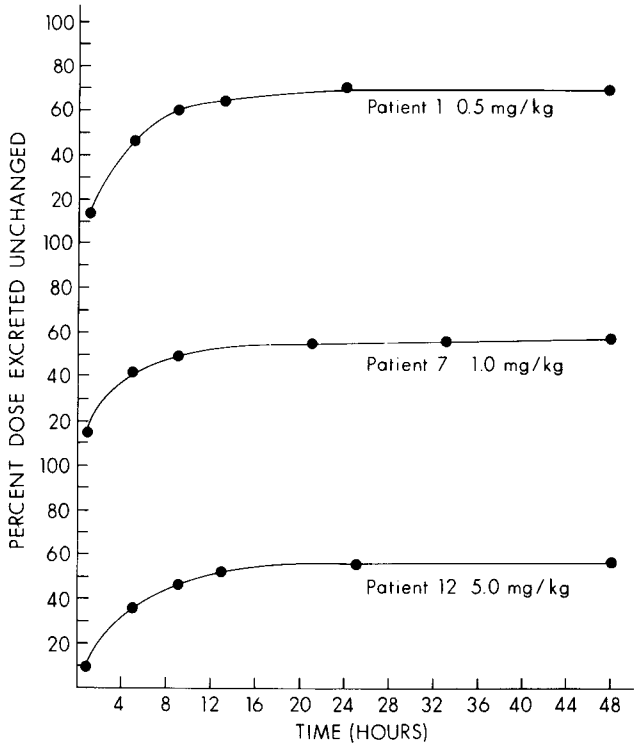


Fig. 4. Representative cumulative urinary excretion curves of acyclovir at 3 different doses.

69%. Of 13 patients 7 had recoveries greater than 60%. Representative plots of urinary excretion rate vs. midpoint plasma concentration are shown in Fig. 2. The correlation coefficients for the data used in each regression are given in Table IV. Cl_R ranged from 48.5 to 245.4 ml/min/1.73 m^2 with a mean of 159 ml/min/1.73 m^2, excluding one patient (No. 11) with a questionable Cl_R value of 559.1. Estimates of Cl_{cr} ranged from 33.1 to 87.9 ml/min/1.73 m^2. There was no significant correlation between either estimated or measured Cl_{cr} and Cl_R. The ratio of Cl_R to Cl_{TOT} was computed. Excluding the unrealistic value of 1.413 (Patient No. 11), the mean value of Cl_R/Cl_{TOT} was 0.629 (range 0.304 to 0.919). There was a good correlation ($r = 0.871$) between this ratio and percent unchanged drug excreted in the urine (Fig. 3).

Representative curves of cumulative urinary excretion of unchanged acyclovir (Fig. 4) show that excretion of the drug occurred largely during the infusion and in the 8 hr following infusion. Analyses of selected urine samples by high-performance liquid chromatography (HPLC) indicated the presence of a metabolite with a retention time (16 min) identical with that of synthetic 9-carboxymethoxymethylguanine. Fractions eluted from the HPLC column, corresponding to the ultraviolet (UV) peak with the retention time of 16 min, were concentrated and analyzed by thin-layer chromatography in 3 solvent systems and by paper chromatography (Table V). The R_f values of the isolated metabolite were virtually identical with those of 9-carboxymethoxymethylguanine and when the isolated metabolite and the synthetic 9-carboxymethoxymethylguanine were eluted with water from the paper chromatogram the respective UV spectra were identical. Quantitation of this metabolite in the individual urines of 5 subjects, by comparing the areas of the UV absorption peak at 280 nm for the metabolite with the corresponding UV peak area for the authentic compound, indicated that 2.4% to 13.6% of the dose was present as the metabolite.

Toxicity. No subject demonstrated any evi-

Table V. *Chromatographic (HPLC, TLC, and paper) and spectral properties of the urinary metabolite of acyclovir and of synthetic 9-carboxymethoxymethylguanine*

	Metabolite	*Synthetic sample*
Retention time (min) (HPLC)	16	16
Solvent system A*, R_f (TLC§)	0.66	0.65
Solvent system B†, R_f (TLC§)	0.28	0.26
Solvent system C‡, R_f (TLC§)	0.34	0.34
Solvent system C‡, R_f (paper‖)	0.34	0.34
λ_{max} (nm)	247	247
Shoulder (nm)	270	270
Ratio 270/247	0.677	0.672

*Ammonium sulfate, 5% and isopropanol (95:5).
†n-Propanol and water (7:3).
‡n-Butanol:acetic acid and water (12:3:5).
§Eastman cellulose sheets with fluorescent indicator.
‖Whatman 3 MM paper.

dence of toxicity either clinically or in laboratory findings. Specifically, hematocrit, white blood count, reticulocyte count, platelets, and parameters of liver and renal function showed no alterations during or after therapy.

Discussion

The kinetic disposition of acyclovir in man after intravenous administration exhibited biphasic characteristics similar to those reported in dogs.[17] Through the use of equation 1, samples collected during the infusion period could be included with the postinfusion data for kinetic model fitting. Parameter estimates derived from the 0.5 and 1.0 mg/kg doses were used to simulate several sets of plasma concentration-time data into which random error was introduced. The results of NONLIN analysis of these simulated data sets were used to optimize the plasma sampling schedule. It was found that the inclusion of samples at 0.25 and 0.75 hr during infusion would more accurately characterize the rapid disposition rate constant (α). Therefore, sample collections at these times were included for the 2.5 and 5.0 mg/kg studies. Besides an obvious dependence on the sampling schedule, the reliable determination of α, V_1, and k_{12} of a drug administered by intravenous infusion is particularly sensitive to deviations from a constant infusion rate. This may be difficult to achieve in the clinical setting unless a precalibrated constant rate infusion pump is used. For Patients No. 4, 8, and 12 low parameter estimates of α and calculated k_{12} values might be explained by departure from a constant input rate during infusion.

The acyclovir t½ was constant over the 0.5 to 5 mg/kg dose range studied. For 13 of the 14 patients studied, plasma t½ ranged from 2.1 to 3.8 hr, nearly identical to the t½ range reported for dogs;[17] 1 patient had a slightly longer t½ (5.0 hr). The calculated values of Cl_{TOT} varied 3-fold (range 117 to 396 ml/min/1.73 m^2), with a trend toward higher clearances at higher doses. This trend may be related to underlying pathology rather than dose. The two subjects with the highest Cl_{TOT} (patients Nos. 9 and 11) had a surgical history of radical nephrectomy. The trend toward larger volumes of distribution at higher doses is consistent with increasing Cl_{TOT} and unchanged t½. Additional studies currently in progress will indicate whether there is a dose-dependent change in acyclovir clearance.

The Cl_{TOT} of Patient No. 11 was the highest reported, but the t½ value was in the same range as that of the other 13 patients. The analysis of plasma and urine data from Patient No. 11 resulted in unexplicable anomalies. This was the only subject whose plasma concentration-time profile was not well fit by the 2-compartment model. His Cl_R/Cl_{TOT} had the unrealistic value of 1.41. If it is assumed that Cl_{TOT} is correct, the Cl_R can be estimated from the linear relationship between percent drug excreted unchanged and Cl_R/Cl_{TOT} (Fig. 3). This estimated Cl_R would be

307 ml/min/1.73 m^2, which is more consistent with the other Cl_R values than the observed value of 559.1 ml/min/1.73 m^2.

The pattern of the cumulative urinary excretion of acyclovir parallels the observed biphasic disposition of the drug from the plasma. The calculated renal clearances of acyclovir, as well as the percent dose excreted unchanged in the urine, varied widely among the 14 subjects. The drug clearances (Patient No. 11 excluded) were substantially higher than the estimated or experimentally determined creatinine clearances, an indication that, in addition to glomerular filtration, tubular secretion contributes to the renal elimination of the drug. Renal function, as assessed by creatinine clearance, and Cl_{TOT} or Cl_R did not correlate. The wide range of Cl_R/Cl_{TOT} ratios (0.304 to 0.919) indicates a high degree of patient variability in renal to nonrenal excretion of the drug and suggests metabolism and other extrarenal factors in the disposition of acyclovir. Intersubject variability may be related to underlying pathology or perhaps drug-drug interactions that could influence the acyclovir disposition. The drug exhibited low plasma protein binding (22% to 33% as determined by ultrafiltration at 37°), suggesting that this mechanism of drug-drug interaction is not important. The incomplete urinary recovery of acyclovir could be attributed in part to the formation of 9-carboxymethoxymethylguanine, which was identified as a minor metabolite in the mouse[23] and other species.[5] This metabolite was found to be present in several representative urine samples analyzed in the present study. Due to the presence of endogenous UV-absorbing constituents in the urine it was not possible to quantitate this metabolite accurately or to determine possible lower concentrations excreted in later urine samples.

Other metabolites could also be formed in man. In the rhesus monkey 8-hydroxy-9-(2-hydroxyethoxymethyl)guanine was identified as a urinary metabolite of acyclovir,* but we could not demonstrate it in this study.

There is also the possibility that acyclovir may, in part, be cleaved to guanine either by phosphorolysis or by the degradation of the acyclic side chain. Subsequently, guanine could be deaminated to xanthine which could be oxidized to uric acid by xanthine oxidase. Such an incorporation of the purine moiety of acyclovir into the large endogenous uric acid pool, and also possibly into GTP, can be demonstrated only with the use of drug radioactively labeled in the guanine moiety. Extrarenal excretion (particularly into the feces) of uric acid[26] as well as of acyclovir and possible metabolites are areas for further investigation. Studies using the ^{14}C-labeled drug are planned in order to define completely the metabolic fate of this drug in man.

Our study demonstrates that, in addition to its high selectivity and low toxicity,[27] acyclovir has a kinetic profile and metabolic disposition which make it an attractive candidate for therapy in a variety of herpes infections. These studies indicate that administration of acyclovir by 1-hr infusions every 8 hr results in the persistence of drug levels which have been found active in vitro against herpes simplex virus types 1 and 2 and varicella zoster virus. Other nucleoside analogues with antiviral activity have shorter t½s or are extensively metabolized[3, 7, 13, 18] and are generally suitable for intravenous administration only when given by continuous infusion.[2, 12, 28] Adenine arabinoside, a drug currently approved for treatment of herpes keratitis and encephalitis, is metabolized to arabinosyl hypoxanthine, which has much lower antiviral activity than the parent compound and, due to its relative insolubility, administration of adenine arabinoside requires large volumes of intravenous fluids and prolonged intravenous infusions (12 hr), a problem for patients with brain edema due to herpes encephalitis.[28] The advantages of acyclovir are greater potency, solubility, and anabolism in the infected cell to an active form.

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